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## REINFORCEMENT MEMBER ASSEMBLY FOR SIDE SILL OF VEHICLE

### Abstract

A reinforcement member assembly for a side sill of a vehicle includes an upper reinforcement member formed in a longitudinal direction of the vehicle and provided inside a side sill of the vehicle; a lower reinforcement member positioned below the upper reinforcement member, including one side in a width direction of the vehicle, which is connected to the upper reinforcement member in the longitudinal direction of the vehicle, provided inside the side sill; and internal reinforcement members connected to a bottom surface of the upper reinforcement member and an upper surface of the lower reinforcement member, respectively, which reduces weight of the vehicle and improves strength of the vehicle against a side collision.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] The present application claims priority to Korean Patent Application No. 10-2024-0028846, filed on Feb. 28, 2024, the entire contents of which is incorporated herein for all purposes by this reference.
BACKGROUND OF PRESENT DISCLOSURE
Field of Present Disclosure
[0002] The present disclosure relates to a reinforcement member assembly for a side sill of a vehicle, which enables weight reduction and improves side collision performance.
Description of Related Art
[0003] Referring to FIGS. **1-3**, a side sill **110** is provided at a lower portion of a side surface of a vehicle in a longitudinal direction of the vehicle.
[0004] The side sill **110** is formed by joining a side sill inner **111** formed convexly toward an interior of the vehicle and a side sill outer **112** formed convexly toward an outside of the vehicle. Upper and lower end portions of the side sill inner **111** and the side sill outer **112** are connected to each other to form a hollow structure in a longitudinal direction of the vehicle, forming a lower structure of the vehicle.
[0005] Meanwhile, a reinforcement member **120** is provided inside the side sill **110** to reinforce rigidity of the side sill **110**. For example, the reinforcement member **120** is manufactured by extruding an aluminum alloy. Since the reinforcement member **120** is made of a different material from those of the side sill inner **111** and the side sill outer **112**, which are typically manufactured from a steel sheet, the reinforcement member **120** may not be connected to the side sill inner **111** or the side sill outer **112** by welding.
[0006] To fasten the reinforcement member **120** to the side sill inner **111** or the side sill outer **112**, brackets **121** and **122** are connected to the reinforcement member **120** using a screw **123** through a flow drill screw (FDS) method, and then the brackets **121** and **122** are connected to the side sill inner **111** or the side sill outer **112** by welding or the like so that the reinforcement member **120** may be provided inside the side sill **110**.
[0007] However, the reinforcement member **120** is manufactured by extruding an expensive aluminum alloy, increasing manufacturing costs of the vehicle. Furthermore, to install the reinforcement member **120**, a process of fastening the brackets **121** and **122** by the FDS method and a process of welding the brackets **121** and **122** to the side sill inner **111** or the side sill outer **112** are discretely performed so that additional production lines should be provided to increase an amount of work as well as production costs of the vehicle. For example, the brackets **121** and **122** are firstly connected to the reinforcement member **120**, and then connected to an assembly of the inner panel **113** of the vehicle such as a center pillar inner panel and the side sill inner **111**, and the side sill outer **112** is finally connected to outer an external side of the assembly.
[0008] Furthermore, since the reinforcement member **120** is manufactured by a extrusion process, a thickness of a cross section of the reinforcement member **120** inevitably becomes thick, and as a result, a gap between a circumference of the reinforcement member **120** and the side sill inner **111** or the side sill outer **112** becomes narrow. This prevents an electrodeposition solution from flowing smoothly during a painting process, causing the electrodeposition solution to clump together in some areas resulting in excessive painting or causing a painting defect in which the electrodeposition solution is not applied to some areas. That is, since a gap between the reinforcement member **120**, the side sill inner **111**, and the side sill outer **112** is narrow as shown in

FIG. **3** and a gap between the reinforcement member **120** and the side sill inner **111** and the side sill outer **112** is narrow as shown in FIG. **4** at both sides and a lower portion of the reinforcement member **120**, the electrodeposition solution may not flow smoothly, and thus the electrodeposition solution is not applied uniformly, causing a painting defect.

[0009] Furthermore, for a vehicle such as a multi-purpose vehicle (MPV), a battery is provided in the vehicle. As the battery is provided, a weight of the vehicle increases, and thus a load that the vehicle should support in the event of a side collision increases. However, since a sliding door is usually provided in the vehicle such as an MPV, a cross-sectional area of the side sill **110** is reduced to install a rail for opening and closing the sliding door, and thus it is difficult to apply the reinforcement member **120** made by extruding the aluminum alloy, and there is a problem in that a bearing capacity against a side collision is reduced.

[0010] As described above, when the bearing capacity against the side collision is weakened, there is a problem in that the battery provided for electrification is damaged due to the side collision.

[0011] The information included in this Background of the present disclosure is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.

BRIEF SUMMARY

[0012] Various aspects of the present disclosure are directed to providing a reinforcement member assembly for a side sill of a vehicle, which may reduce a weight compared with a reinforcement member of an aluminum extrusion material and exhibit comparable collision performance.

[0013] An exemplary embodiment of the present disclosure is also directed to providing a reinforcement member assembly for a side sill of a vehicle, which is easily provided inside the side sill to reduce manufacturing costs and does not hinder a flow of an electrodeposition solution during a painting process.

[0014] Other objects and advantages of the present disclosure may be understood by the following description and become apparent with reference to the exemplary embodiments of the present disclosure. Also, it is obvious to those skilled in the art to which the present disclosure pertains that the objects and advantages of the present disclosure may be realized by the means as claimed and combinations thereof.

[0015] In accordance with an exemplary embodiment of the present disclosure, there is provided a reinforcement member assembly for a side sill of a vehicle, which includes an upper reinforcement member formed in a longitudinal direction of the vehicle and provided inside the side sill of the vehicle; a lower reinforcement member positioned below the upper reinforcement member, including one side in a width direction of the vehicle, which is connected to the upper reinforcement member in the longitudinal direction of the vehicle, provided inside the side sill; and an internal reinforcement member connected to a bottom surface of the upper reinforcement member and an upper surface of the lower reinforcement member, respectively.

[0016] The upper reinforcement member may include an upper surface and a side surface bent downwardly from the upper surface, and the lower reinforcement member may include a lower surface positioned below the upper surface of the upper reinforcement member and a side surface bent upwards from the lower surface and connected to the side surface of the upper reinforcement member.

[0017] The upper reinforcement member may include bead portions formed on the upper surface to be higher than the upper surface in a width direction of the vehicle, and the bead portions may be formed at intervals in the longitudinal direction of the vehicle.

[0018] A through-hole passing through the bead portion may be formed in the bead portion.

[0019] The lower reinforcement member may include bead portions formed on the lower surface to be lower than the lower surface in the width direction of the vehicle, and the bead portions may be formed at intervals in the longitudinal direction of the vehicle.

[0020] A through-hole may be formed on each of the lower surface and the side surface of the upper reinforcement member.
[0021] The internal reinforcement member may be provided as a plurality of internal reinforcement members and disposed at intervals in the longitudinal direction of the vehicle.
[0022] The internal reinforcement member may include an upper bonding portion connected to the bottom surface of the upper reinforcement member, a lower bonding portion disposed at both end portions of the upper bonding portion in the longitudinal direction of the vehicle and connected to the upper surface of the lower reinforcement member, and a connection portion which connects the upper bonding portion and the lower bonding portion.
[0023] A bead portion may be formed in the upper bonding portion to be lower than the upper bonding portion in the width direction of the vehicle.
[0024] A bead portion may be formed on an upper surface of the upper reinforcement member to be higher than the upper surface in the width direction of the vehicle, and the bead portion of the internal reinforcement member may be positioned below the bead portion.
[0025] A through-hole may be formed in the bead portion.
[0026] The reinforcement member assembly may further include a connection member whose front and rear end portions are connected to internal reinforcement members adjacent to each other in the longitudinal direction of the vehicle.
[0027] A bead portion may be formed in the connection member in the width direction of the vehicle.
[0028] A bead portion may be formed on an upper surface of the upper reinforcement member to be higher than the upper surface in the width direction of the vehicle, and the bead portion of the connection member may be positioned below the bead portion of the upper reinforcement member.
[0029] The connection member may be disposed between the upper reinforcement member and the internal reinforcement member and connected thereto.
[0030] The connection member may include a first internal reinforcement member disposed as a plurality of first internal reinforcement members at intervals in the longitudinal direction of the vehicle, and a second internal reinforcement member disposed with an interval from a first internal reinforcement member, among the first internal reinforcement members, disposed at a frontmost or rearmost position in the longitudinal direction of the vehicle.
[0031] The upper reinforcement member and the lower reinforcement member may be connected to a side sill inner of the side sill.
[0032] The reinforcement member assembly may further include an extension member connected to a front or a rear end portion of the lower reinforcement member.
[0033] A second connection member connects the first internal reinforcement member and the second internal reinforcement member, and the second connection member may be formed to have a different length from the first connection member.
[0034] The extension member may be formed to have substantially the same cross-sectional shape as the lower reinforcement member.
[0035] The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0036] FIG. **1** is a perspective view exemplarily illustrating a state in which a reinforcement member is provided inside a side sill according to the related art.

[0037] FIG. **2** is a perspective view exemplarily illustrating a state in which a bracket is provided on the reinforcement member according to the related art.
[0038] FIG. **3** is a cross-sectional view exemplarily illustrating a center pillar portion of a vehicle in the side sill according to the related art.
[0039] FIG. **4** is a cross-sectional view exemplarily illustrating a center pillar of the vehicle in the side sill according to the related art when viewed from the rear.
[0040] FIG. **5** is a partial perspective view exemplarily illustrating a state in which a reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure is provided at a side sill internal.
[0041] FIG. **6** is an exploded perspective view of FIG. **5** illustrating a state in which the reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure is separated from the side sill internal.
[0042] FIG. **7** is a perspective view exemplarily illustrating the reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure.
[0043] FIG. **8** is an exploded perspective view of FIG. **7** illustrating the reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure.
[0044] FIG. **9** is a bottom perspective view of FIG. **5** illustrating the reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure.
[0045] FIG. **10** is a side view of FIG. **5** illustrating the reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure.
[0046] FIG. **11** is a cross-sectional view taken along line I-I of FIG. **10**.
[0047] It may be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as included herein, including, for example, specific dimensions, orientations, locations, and shapes locations, and shapes will be determined in part by the particularly intended application and use environment.
[0048] In the figures, reference numbers refer to the same or equivalent portions of the present disclosure throughout the several figures of the drawing.

## DETAILED DESCRIPTION

[0049] Reference will now be made in detail to various embodiments of the present disclosure(s), examples of which are illustrated in the accompanying drawings and described below. While the present disclosure(s) will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that the present description is not intended to limit the present disclosure(s) to those exemplary embodiments of the present disclosure. On the other hand, the present disclosure(s) is/are intended to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims.
[0050] Hereinafter, a reinforcement member assembly for a side sill according to an exemplary embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.
[0051] A reinforcement member assembly **20** for a side sill according to an exemplary embodiment of the present disclosure may be provided inside a side sill **10** of a vehicle to improve a rigidity of the side sill **10**.
[0052] The side sill **10** may be formed by bonding a side sill inner **11** formed convexly toward an interior of the vehicle and a side sill outer **12** formed convexly toward an outside of the vehicle.
[0053] The reinforcement member assembly **20** for a side sill may be provided inside the side sill **10** to improve the rigidity of the side sill **10**.
[0054] The reinforcement member assembly **20** for a side sill according to an exemplary embodiment of the present disclosure may include an upper reinforcement member **21** formed in a

longitudinal direction of the vehicle and provided inside the side sill **10** of the vehicle; a lower reinforcement member **22** positioned below the upper reinforcement member **21**, including one side in a width direction of the vehicle, which is connected to the upper reinforcement member **21** in the longitudinal direction of the vehicle, provided inside the side sill **10**; and internal reinforcement members **23** and **24** connected to a bottom surface of the upper reinforcement member **21** and an upper surface of the lower reinforcement member **22**, respectively.

[0055] The upper reinforcement member **21** may be formed in the longitudinal direction of the vehicle.

[0056] The upper reinforcement member **21** may include an upper surface **21***a* formed in the longitudinal direction of the vehicle and a side surface **21***b* extending downwardly from a side surface of the upper surface **21***a*. The upper reinforcement member **21** may include a bent cross section by the upper surface **21***a* and the side surface **21***b*, which is formed continuously in the longitudinal direction of the vehicle, so that a cross section of an upper side of the reinforcement member assembly **20** for a side sill is formed.

[0057] A bead portion **21***aa* may be formed on the upper surface **21***a* to be higher than the upper surface **21***a*. Since the bead portion **21***aa* is formed higher than the upper surface **21***a* in the width direction of the vehicle, the bead portion **21***aa* improves a side collision load that the upper surface **21***a* may experience in the event of a side collision of the vehicle. The bead portion **21***aa* may be formed as a plurality of bead portions **21***aa* on the upper surface **21***a* at intervals therebetween in the longitudinal direction of the vehicle. Since the bead portion **21***aa* formed on the upper surface **21***a* is formed to be higher than the upper surface **21***a*, the bead portion **21***aa* may include a convex shape upwards from the upper surface **21***a*. Since the bead portion **21***aa* supports the collision load input to the upper surface **21***a* in the width direction of the vehicle, rigidity of the reinforcement member assembly **20** for a side sill may be improved. The bead portion **21***aa* may increase a cross-sectional coefficient of the upper surface **21***a* to improve the rigidity. The bead portion **21***aa* may be basically formed in the width direction of the vehicle, and alternatively, some bead portions **21***aa* may be additionally formed to protrude in the longitudinal direction of the vehicle.

[0058] A through-hole **21***ab* may be formed in the bead portion **21***aa*. Since the through-hole **21***ab* is formed to pass through the bead portion **21***aa*, when a vehicle body of the vehicle is electrodeposition-painted, an electrodeposition solution may freely flow through the through-hole **21***ab* to facilitate the application of the electrodeposition solution, preventing the electrodeposition solution from clumping in some areas or the electrodeposition solution from not being flown.

[0059] The lower reinforcement member **22** may be positioned below the upper reinforcement member **21**, and one side of the lower reinforcement member **22** may be connected to the upper reinforcement member **21** in the longitudinal direction of the vehicle.

[0060] The lower reinforcement member **22** may be provided inside the side sill **10** in a state of being connected to the upper reinforcement member **21**, and thus an assembly of the upper reinforcement member **21** and the lower reinforcement member **22** may form an integrated structural body.

[0061] The lower reinforcement member **22** may include a lower surface **22***a* positioned below the upper surface **21***a*, and a side surface **22***b* bent upwards from the lower surface **22***a* and connected to the side surface **21***b* of the upper reinforcement member **21**. The lower reinforcement member **22** may form a cross section of a lower portion in the reinforcement member assembly **20** for a side seal by forming a bent cross section by the lower surface **22***a* and the side surface **22***b*.

[0062] The lower reinforcement member **22** may be connected to the upper reinforcement member **21** so that the assembly thereof becomes a state in which a U-shape is rotated at 90 degrees. Accordingly, the rigidity of the side sill **10** may be reinforced by the cross-sectional structure formed by the upper reinforcement member **21** and the lower reinforcement member **22**. The assembly of the lower reinforcement member **22** and the upper reinforcement member **21** may include an open cross section facing the outside of the vehicle by rotating a ‘U’-shaped cross section so that the side collision load input in the width direction of the vehicle is supported. The side surface **22***b* of the lower reinforcement member **22** may be connected to the side surface **21***b* of the upper reinforcement member **21** through welding, for example, spot welding.

[0063] Bead portions **22***aa* may be formed on the lower surface **22***a* of the lower reinforcement member **22**. The bead portion **22***aa* may be formed on the lower surface **22***a* to be lower than the lower surface **22***a* in the width direction of the vehicle. The bead portions **22***aa* may be formed on the lower surface **22***a* at intervals therebetween in the longitudinal direction of the vehicle to increase the collision load that the lower surface **22***a* may support in the event of a side collision of the vehicle, improving rigidity of the lower surface **22***a*. The bead portion **22***aa* may be formed to extend to the side surface **22***b*.

[0064] A through-hole **22***ab* passing through the lower surface **22***a* may be formed on the lower surface **22***a* of the lower reinforcement member **22**. Furthermore, a through-hole **22***ba* passing through the side surface **22***b* may be also formed on the side surface **22***b*. The through-holes **22***ab* and **22***ba* formed on the lower surface **22***a* and the side surface **22***b*, respectively, may allow the electrodeposition solution to flow smoothly during the painting process and secure that the electrodeposition solution is uniformly applied.

[0065] The internal reinforcement members **23** and **24** may be connected to the bottom surface of the upper reinforcement member **21** and the upper surface of the lower reinforcement member **22**, respectively. The internal reinforcement members **23** and **24** may be provided as a plurality of internal reinforcement members **23** and **24** disposed at intervals in the longitudinal direction of the vehicle. Since the upper reinforcement member **21** and the lower reinforcement member **22** are connected to each other through the internal reinforcement members **23** and **24**, in the event of a side collision, the upper reinforcement member **21** and the lower reinforcement member **22** is configured to prevent the upper reinforcement member **21** and the lower reinforcement member **22** from being separated. That is, since the upper reinforcement member **21** and the lower reinforcement member **22** are connected by the internal reinforcement members **23** and **24** in the longitudinal direction of the vehicle, when a side collision of the vehicle occurs, the upper reinforcement member **21** and the lower reinforcement member **22** may prevent a phenomenon in which the upper reinforcement member **21** and the lower reinforcement member **22** are separated from each other.

[0066] Furthermore, the internal reinforcement members **23** and **24** may also be configured to directly support the side collision load. Since the internal reinforcement members **23** and **24** are formed as an open cross section in the width direction of the vehicle, the internal reinforcement members **23** and **24** may also support the side collision load.

[0067] The internal reinforcement members **23** and **24** may include a plurality of first internal reinforcement members **23** disposed in the longitudinal direction of the vehicle and a plurality of second internal reinforcement members **24** disposed in front of or behind the first internal reinforcement members **23**.

[0068] The first internal reinforcement member **23** may include an upper bonding portion **23***a* connected to the bottom surface of the upper reinforcement member **21**, a lower bonding portion **23***b* disposed at both end portions of the upper bonding portion **23***a* in the longitudinal direction of the vehicle and connected to the upper surface of the lower reinforcement member **22**, and a connection portion **23***c* connecting the upper bonding portion **23***a* and the lower bonding portion **23***b*.

[0069] The upper bonding portion **23***a* may be connected to the bottom surface of the upper reinforcement member **21**, that is, a bottom surface of the upper surface **21***a*. The upper bonding portion **23***a* may be formed to have a predetermined length in the longitudinal direction of the vehicle and may be connected to the upper surface **21***a*, fastening the first internal reinforcement member **23** to the upper reinforcement member **21**.

[0070] A bead portion **23***aa* may be formed in the upper bonding portion **23***a* to be lower than the upper bonding portion **23**_a_ in the width direction of the vehicle. The bead portion **23**_aa_ formed in the upper bonding portion **23**_a_ may be positioned below the bead portion **21**_aa_ formed in the upper reinforcement member **21** when the first internal reinforcement member **23** is connected to the upper reinforcement member **21** (see FIG. **9**). While the bead portion **21**_aa_ of the upper reinforcement member **21** includes a convex shape, the bead portion **23**_aa_ of the upper bonding portion **23**_a_ may include a concave shape, and thus when the upper bonding portion **23**_a_ is connected to the upper reinforcement member **21**, the bead portions **21**_aa_ and **23**_a_ may form a cylindrical structure in the width direction of the vehicle. This may support a side collision load in the event of a side collision.

[0071] In the first internal reinforcement member **23**, a through-hole **23**_ab_ may be formed in the bead portion **23**_aa_ to allow the electrodeposition solution to flow smoothly during the painting process.

[0072] Similar to the first internal reinforcement member **23**, the second internal reinforcement member **24** may include an upper bonding portion **24**_a_ connected to the bottom surface of the upper reinforcement member **21**, a lower bonding portion **24**_b_ disposed at both end portions of the upper bonding portion **24**_a_ in the longitudinal direction of the vehicle and connected to the upper surface of the lower reinforcement member **22**, and a connection portion **24**_c_ connecting the upper bonding portion **24**_a_ and the lower bonding portion **24**_b_.

[0073] The second internal reinforcement member **24** may be connected to the upper reinforcement member **21** and the lower reinforcement member **22** by welding, and similar to the first internal reinforcement member **23**, the second internal reinforcement member **24** may prevent the upper reinforcement member **21** and the lower reinforcement member **22** from being separated in the event of a side collision.

[0074] The second internal reinforcement member **24** is configured to connect the upper reinforcement member **21** and the lower reinforcement member **22** at a front side or a rear side of the first internal reinforcement member **23** disposed as a plurality of first internal reinforcement members **23**. That is, the second internal reinforcement member **24** may be positioned in front of the first internal reinforcement member **23**, which is positioned at the foremost position among the first internal reinforcement members **23**, in the longitudinal direction of the vehicle or positioned behind the first internal reinforcement member **23**, which is positioned at the rearmost position among the first internal reinforcement members **23**, in the longitudinal direction of the vehicle.

[0075] Connection members **25** and **26** may be disposed in the longitudinal direction of the vehicle, and front and rear end portions of the connection members **25** and **26** may be connected to the upper bonding portions **23**_a_ and **24**_a_ of the internal reinforcement members **23** and **24**.

[0076] In the internal reinforcement members **23** and **24**, the upper bonding portions **23**_a_ and **24**_a_ may be connected to the bottom surface of the upper reinforcement member **21**, and thus the connection members **25** and **26** may be disposed between the upper reinforcement member **21** and the internal reinforcement members **23** and **24**.

[0077] The connection members **25** and **26** may include a first connection member **25** connecting the two first internal reinforcement members **23** positioned adjacent to each other in the longitudinal direction of the vehicle, and a second connection member **26** connecting the first internal reinforcement member **23** and the second internal reinforcement member **24**.

[0078] The first connection member **25** may connect the two internal reinforcement members **23** and **24** disposed in the longitudinal direction of the vehicle, and may prevent rotation or deformation of the internal reinforcement members **23** and **24** in the event of a side collision of the vehicle, preventing the internal reinforcement members **23** and **24** from being separated from the upper reinforcement member **21** and the lower reinforcement member **22**.

[0079] The first connection member **25** may be disposed between the upper reinforcement member **21** and the first internal reinforcement member **23** to connect two internal reinforcement members **23** and **24** disposed adjacent to each other in the longitudinal direction of the vehicle.

[0080] The first connection member **25** may be connected to the upper reinforcement member **21** and the first internal reinforcement member **23** by welding.
[0081] A bead portion **25***a* may be formed in the first connection member **25** in the width direction of the vehicle. The bead portion **25***a* may be formed to be concave downward and may be positioned below the bead portion **21***aa* formed in the upper reinforcement member **21** (see FIG. **9**). When the first connection member **25** is connected to the upper reinforcement member **21**, the bead portions **21***aa* and **25***a* formed in the upper reinforcement member **21** and the first connection member **25** may form a cylindrical structure in the width direction of the vehicle, supporting a side collision load in the event of a side collision.
[0082] A through-hole **25***b* may be formed in the first connection member **25** to facilitate the flow of the electrodeposition solution during the painting process.
[0083] The second connection member **26** may be formed in a shape similar to a shape of the first connection member **25**. The second connection member **26** may be formed to have a different length from the first connection member **25**. Since the second internal reinforcement member **24** is formed in a different form from the first internal reinforcement member **23**, and a distance between the second internal reinforcement member **24** and the first internal reinforcement member **23** adjacent thereto may be different from a distance at which the first internal reinforcement members **23** are disposed, the second connection member **26** may be formed to have a different length from the first connection member **25**.
[0084] Similar to the first connection member **25**, a bead portion **26***a* may be formed in the second connection member **26** in the width direction of the vehicle. Since the bead portion **26***a* is also positioned below the bead portion **21***aa* formed in the upper reinforcement member **21** and is formed to be concave downward, the bead portion **21***aa* and the bead portion **26***a* may also form a cylindrical structure in the width direction of the vehicle, supporting a side collision load in the event of a side collision.
[0085] A through-hole **26***b* may be formed to pass through the second connection member **26** to facilitate the flow of the electrodeposition solution during the painting process.
[0086] The second connection member **26** may be also connected to the bottom surface of the upper reinforcement member **21**, and the upper bonding portion **23***a* of the first internal reinforcement member **23** and the upper bonding portion **24***a* of the second internal reinforcement member **24** may be connected to a bottom surface of the second connection member **26**. Thus, the second connection member **26** may be disposed between the upper reinforcement member **21** and the upper bonding portion **23***a* or between the upper reinforcement member **21** and the upper bonding portion **24***a*.
[0087] The second connection member **26** may be connected to the upper reinforcement member **21**, the first internal reinforcement member **23**, and the second internal reinforcement member **24** by welding.
[0088] An extension member **27** may be connected to a front end portion or a rear end portion of the lower reinforcement member **22**. The extension member **27** may be formed to have substantially the same cross-sectional shape as the lower reinforcement member **22** and is configured to extend the lower reinforcement member **22**.
[0089] The reinforcement member assembly **20** for a side sill according to an exemplary embodiment of the present disclosure may be provided inside the side sill **10** of the vehicle. The upper reinforcement member **21** and the lower reinforcement member **22** may be connected to the side sill inner **11** of the side sill **10** by welding. The reinforcement member assembly **20** for a side sill may be assembled in advance, and then during an assembly process of the side sill **10**, the reinforcement member assembly **20** for a side sill may be connected to an assembly of the side sill inner **11** and the inner panel **13** of the vehicle body.
[0090] The reinforcement member assembly **20** of a side sill may be connected to the side sill inner **11**, and then the side sill outer **12** may be connected to an external side of the assembly of the side

sill inner **11** and the reinforcement member assembly **20** of a side sill. Thus, the side sill **10** provided with the reinforcement member assembly **20** of a side sill may be formed therein.
[0091] Since the reinforcement member assembly **20** of a side sill is connected to the side sill inner **11** by welding, an amount of work required may be reduced compared to when applying the reinforcement member **120** made of an aluminum extrusion material, reducing the time and cost required for production of the vehicle.
[0092] Furthermore, since each member forming the reinforcement member assembly **20** of a side sill is manufactured by welding metal plates such as steel sheets, it is possible to reduce a weight compared to the conventional thick aluminum extrusion material and allow the flow of the electrodeposition solution inside to become smoother.
[0093] An operation of the reinforcement member assembly **20** of a side sill according to an exemplary embodiment of the present disclosure including the above configuration will be described below.
[0094] The reinforcement member assembly **20** for a side sill according to an exemplary embodiment of the present disclosure may be provided inside the side sill **10**.
[0095] When a side collision of the vehicle occurs, a collision load may be applied to the side sill outer **12**.
[0096] The collision load input to the side sill outer **12** may be transmitted to the reinforcement member assembly **20** for the side sill **10**.
[0097] Since the upper reinforcement member **21** and the lower reinforcement member **22** are connected to each other, and a cross section thereof may be formed to face outer an external side in the width direction of the vehicle, the upper surface **21***a* of the upper reinforcement member **21** and the lower surface **22***a* of the lower reinforcement member **22** may support the collision load being applied thereto.
[0098] Furthermore, in the event of a side collision, since the first internal reinforcement member **23** and the second internal reinforcement member **24** prevent the upper reinforcement member **21** and the lower reinforcement member **22** from being spread or separated, the upper reinforcement member **21** and the lower reinforcement member **22** may firmly support the collision load.
[0099] Furthermore, since the bead portions **21***aa*, **22***aa*, **23***aa*, **25***a*, and **26***a* are formed in the upper reinforcement member **21**, the lower reinforcement member **22**, the first internal reinforcement member **23**, the first connection member **25**, and the second connection member **26**, respectively, the bead portions **21***aa*, **22***aa*, **23***aa*, **25***a*, and **26***a* may firmly support a load from a side collision.
[0100] According to a reinforcement member assembly for a side sill of a vehicle of the present disclosure, which includes the above-described configuration, the reinforcement member assembly for a side sill may support a collision load input through a side surface of the vehicle so that a phenomenon in which the side sill is excessively deformed may be prevented.
[0101] As the deformation of the side sill is reduced in the event of a side collision, deformation of a compartment due to the side collision may be reduced so that a survival space for occupants may be expanded.
[0102] Furthermore, in eco-friendly vehicles, such as electric vehicles, in which batteries are provided below floors of the vehicles, a bearing capacity against a side collision may be increased so that damage to the battery due to side collision may be prevented.
[0103] For installation of a sliding door in a vehicle such as a multi-purpose vehicle (MVP) to which the sliding door is applied, even when a cross-sectional size of the side sill becomes small, a collision load may be sufficiently supported.
[0104] Since a thin metal plate is used inside the side sill instead of a thick aluminum extrusion material, a weight reduction of the reinforcement member assembly for a side sill may be achieved compared with the reinforcement member of the aluminum extrusion material.
[0105] Furthermore, as a gap between the reinforcement member assembly for a side sill, a side sill

internal, and a side sill outer is increased, a flow of an electrodeposition solution may be facilitated. Furthermore, since a through-hole through which the electrodeposition solution flows is formed, a phenomenon of the electrodeposition solution clumping or not being applied during a painting process is prevented so that painting quality may be improved and corrosion resistance may be improved.

[0106] Furthermore, since the reinforcement member assembly for a side sill is connected to the side sill inner in a single process, an amount of work required may be reduced and there may be no need to use expensive aluminum material for the reinforcement member so that production costs may be reduced.

[0107] In an exemplary embodiment of the present disclosure, the vehicle may be referred to as being based on a concept including various means of transportation. In some cases, the vehicle may be interpreted as being based on a concept including not only various means of land transportation, such as cars, motorcycles, trucks, and buses, that drive on roads but also various means of transportation such as airplanes, drones, ships, etc.

[0108] For convenience in explanation and accurate definition in the appended claims, the terms “upper”, “lower”, “inner”, “outer”, “up”, “down”, “upwards”, “downwards”, “front”, “rear”, “back”, “inside”, “outside”, “inwardly”, “outwardly”, “interior”, “exterior”, “internal”, “external”, “forwards”, and “backwards” are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood that the term “connect” or its derivatives refer both to direct and indirect connection.

[0109] The term “and/or” may include a combination of a plurality of related listed items or any of a plurality of related listed items. For example, “A and/or B” includes all three cases such as “A”, “B”, and “A and B”.

[0110] In exemplary embodiments of the present disclosure, “at least one of A and B” may refer to “at least one of A or B” or “at least one of combinations of at least one of A and B”. Furthermore, “one or more of A and B” may refer to “one or more of A or B” or “one or more of combinations of one or more of A and B”.

[0111] In the present specification, unless stated otherwise, a singular expression includes a plural expression unless the context clearly indicates otherwise.

[0112] In the exemplary embodiment of the present disclosure, it should be understood that a term such as “include” or “have” is directed to designate that the features, numbers, steps, operations, elements, parts, or combinations thereof described in the specification are present, and does not preclude the possibility of addition or presence of one or more other features, numbers, steps, operations, elements, parts, or combinations thereof.

[0113] According to an exemplary embodiment of the present disclosure, components may be combined with each other to be implemented as one, or some components may be omitted.

[0114] The foregoing descriptions of specific exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described in order to explain certain principles of the invention and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

## Claims

**1**. A reinforcement member assembly for a side sill of a vehicle, the reinforcement member assembly comprising: an upper reinforcement member formed in a longitudinal direction of the

vehicle and provided inside the side sill of the vehicle; a lower reinforcement member positioned below the upper reinforcement member, including one side in a width direction of the vehicle, which is connected to the upper reinforcement member in the longitudinal direction of the vehicle, provided inside the side sill; and an internal reinforcement member connected to a bottom surface of the upper reinforcement member and an upper surface of the lower reinforcement member, respectively.
**2**. The reinforcement member assembly of claim 1, wherein the upper reinforcement member includes an upper surface and a side surface bent downwardly from the upper surface, and wherein the lower reinforcement member includes a lower surface positioned below the upper surface of the upper reinforcement member and a side surface bent upwards from the lower surface and connected to the side surface of the upper reinforcement member.
**3**. The reinforcement member assembly of claim 2, wherein the upper reinforcement member includes bead portions formed on the upper surface of the upper reinforcement member to be higher than the upper surface of the upper reinforcement member in the width direction of the vehicle, and wherein the bead portions are formed at intervals in the longitudinal direction of the vehicle.
**4**. The reinforcement member assembly of claim 3, wherein a through-hole passing through each bead portion is formed in each bead portion of the upper reinforcement member.
**5**. The reinforcement member assembly of claim 2, wherein the lower reinforcement member includes bead portions formed on the lower surface of the lower reinforcement member to be lower than the lower surface of the lower reinforcement member in the width direction of the vehicle, and wherein the bead portions of the lower reinforcement member are formed at intervals in the longitudinal direction of the vehicle.
**6**. The reinforcement member assembly of claim 5, wherein a through-hole passing is formed on each of the lower surface of the lower reinforcement member and the side surface of the upper reinforcement member.
**7**. The reinforcement member assembly of claim 1, wherein the internal reinforcement member is in plural and the plurality of internal reinforcement members are disposed at intervals in the longitudinal direction of the vehicle.
**8**. The reinforcement member assembly of claim 1, wherein the internal reinforcement member includes: an upper bonding portion connected to the bottom surface of the upper reinforcement member; a lower bonding portion disposed at first and second end portions of the upper bonding portion in the longitudinal direction of the vehicle and connected to the upper surface of the lower reinforcement member; and a connection portion which connects the upper bonding portion and the lower bonding portion.
**9**. The reinforcement member assembly of claim 8, wherein a bead portion of the internal reinforcement is formed in the upper bonding portion to be lower than the upper bonding portion in the width direction of the vehicle.
**10**. The reinforcement member assembly of claim 9, wherein a bead portion of the upper reinforcement member is formed on an upper surface of the upper reinforcement member to be higher than the upper surface of the upper reinforcement member in the width direction of the vehicle, and wherein the bead portion of the internal reinforcement member is positioned below the bead portion of the upper reinforcement member.
**11**. The reinforcement member assembly of claim 9, wherein a through-hole is formed in the bead portion of the upper reinforcement member.
**12**. The reinforcement member assembly of claim 1, wherein the internal reinforcement is in plural, and wherein the reinforcement member assembly further includes: a first connection member whose front and rear end portions are connected to the plurality of internal reinforcement members adjacent to each other in the longitudinal direction of the vehicle.
**13**. The reinforcement member assembly of claim 12, wherein a bead portion is formed in the first

connection member in the width direction of the vehicle.
**14**. The reinforcement member assembly of claim 13, wherein a bead portion is formed on an upper surface of the upper reinforcement member to be higher than the upper surface of the upper reinforcement member in the width direction of the vehicle, and wherein the bead portion of the first connection member is positioned below the bead portion of the upper reinforcement member.
**15**. The reinforcement member assembly of claim 12, wherein the first connection member is disposed between the upper reinforcement member and the internal reinforcement member and connected thereto.
**16**. The reinforcement member assembly of claim 15, wherein the internal reinforcement member includes: a plurality of first internal reinforcement members disposed at intervals in the longitudinal direction of the vehicle; and a second internal reinforcement member disposed with an interval from a first internal reinforcement member, among the first internal reinforcement members, disposed at a frontmost or rearmost position in the longitudinal direction of the vehicle.
**17**. The reinforcement member assembly of claim 1, wherein the upper reinforcement member and the lower reinforcement member are connected to a side sill inner of the side sill.
**18**. The reinforcement member assembly of claim 1, further including: an extension member connected to a front end portion or a rear end portion of the lower reinforcement member.
**19**. The reinforcement member assembly of claim 16, wherein the internal reinforcement member includes a first internal reinforcement member and a second internal reinforcement member, wherein a second connection member connects the first internal reinforcement member and the second internal reinforcement member, and wherein the second connection member is formed to have a different length from a first connection member.
**20**. The reinforcement member assembly of claim 18, wherein the extension member is formed to have substantially the same cross-sectional shape as the lower reinforcement member.